CHAPTER 1 · CONCEPT 2

Variables — What You Change vs What You Observe

Every good experiment separates the one thing you change from the thing you watch — and keeps everything else the same.

The three kinds of variables

A **variable** is anything in an experiment that can change or be changed. To make sense of an experiment, you must sort variables into three groups.

Variable	What it is	Puri example
Changed (independent)	The one thing you deliberately change	Oil temperature
Measured (dependent)	What you observe/measure as a result	Time to puff (seconds)
Kept same (controlled)	Everything else you hold constant	Dough thickness, size, way it is dropped

> **One-line rule:** change **one** thing, measure **one** thing, keep **everything else** the same.

What can we observe or measure?

- **Yes / No answers:** Did the puri puff up? (qualitative)
- **A number we can measure:** How many seconds did it take? (quantitative)

> **Tip:** the variable you *change* goes "in"; the variable you *measure* comes "out". If the changed variable affects the measured one, you have found a relationship.

Where students lose marks

> **Trap 1 — Changing two things at once.** If you change oil temperature *and* dough thickness together, you can't tell which one caused the difference.

> **Trap 2 — Confusing changed and measured.** The thing you *set* is the changed variable; the thing you *read off* at the end is the measured variable.

> **Trap 3 — Forgetting controlled variables.** Marks are given for naming the things kept the same, not just the one changed.

Model answers — write them like this

Example A. A student tests if warmer water dissolves sugar faster. Name the three kinds of variable.

Changed: temperature of the water.

Measured: time taken for the sugar to dissolve (seconds).

Kept same: amount of sugar, amount of water, stirring.

∴ Change temperature, measure time, keep sugar/water/stirring the same.

Example B. "Plants given more sunlight grow taller." What is being changed and what is measured?

Step 1: The thing set by the experimenter = amount of sunlight (changed).

Step 2: The result read off = height of the plant (measured).

∴ Changed = sunlight; Measured = plant height.

Ready to practise? → Open the worksheet